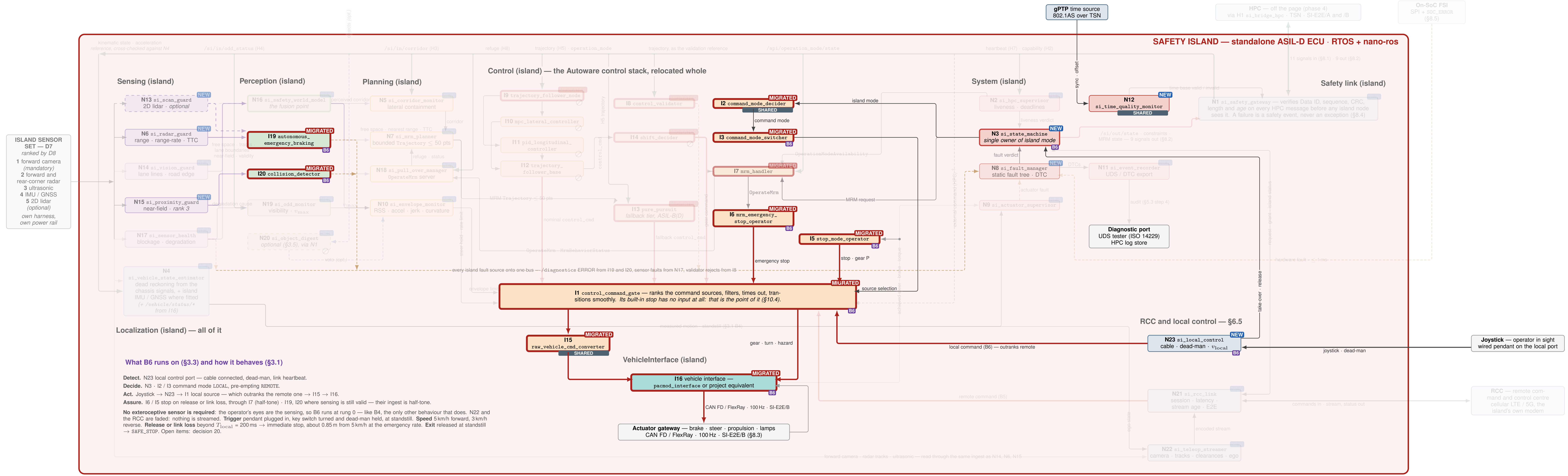
Phase 5 · B6 — Manual joystick control without streamed sensing

An operator in sight of the vehicle drives it with a joystick on the island's local control port — by cable; a radio link is deferred to the next design — with no sensor stream; the island holds the speed to a creep and stops on release or on loss of the link.
The phase-5 drawing, unchanged: what B6 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Stage-1 detailed design: §6.6, entered from standstill.*



SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
gPTP time source
802.1AS over TSN
sync · offset
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
11 signals in (§8.1) · 9 out (§8.2)
objects (opt.)
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
N17 si_sensor_health
blockage · degradation
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_emergency_braking
MIGRATED
B6
I20 collision_detector
N19 si_odd_monitor
visibility · v_{max}
N20 si_object_digest
optional (§3.5), via N1
veto (opt.)
free space · tracks · lane boundaries · near-field · validity
degradation cause
Planning (island)
N5 si_corridor_monitor
lateral containment
perceived corridor
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
v_{max}
N4 si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
I8 control_validator
I14 shift_decider
I13 pure_pursuit
fallback tier, ASIL-B(D)
H5 trajectory
control_cmd
MRM Trajectory ≤ 50 pts
nominal control_cmd
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
steer hold · rate limit
envelope limit
I2 command_mode_decider
SHARED
command mode
I3 command_mode_switcher
island mode
OperationModeAvailability
I7 mrm_handler
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
I5 stop_mode_operator
emergency stop
stop · gear P
source selection
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
measured motion · standstill (§3.1 B4)
external command (HPC) †
achieved steer · brake · torque
I15 raw_vehicle_cmd_converter
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
fault verdict
N8 si_fault_manager
static fault tree · DTC
DTCs
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
actuator fault
N9 si_actuator_supervisor
Diagnostic port
UDS tester (ISO 14229)
HPC log store
N12 si_time_quality_monitor
time base valid / invalid
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
hardware fault · ≤ 1 ms
request · grant · island status
take-over · release
RCC and local control — §6.5
N23 si_local_control
cable · dead-man · v_{local}
local command (B6) — outranks remote
joystick · dead-man
Joystick — operator in sight
wired pendant on the local port
remote command (B5)
N21 si_rcc_link
session · latency · stream age · E2E
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
encoded stream
ego state
N22 si_teleop_streamer
camera · tracks · clearances · ego
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

What B6 runs on (§3.3) and how it behaves (§3.1)

Detect. N23 local control port — cable connected, dead-man, link heartbeat.

Decide. N3 · I2 / I3 command mode `LOCAL`, pre-empting `REMOTE`.

Act. Joystick → N23 → I1 local source — which outranks the remote one → I15 → I16.

Assure. I6 / I5 stop on release or link loss, through I7 (half-tone) · I19, I20 where sensing is still valid — their ingest is half-tone.

No exteroceptive sensor is required: the operator's eyes are the sensing, so B6 runs at rung 0 — like B4, the only other behaviour that does. N22 and the RCC are faded: nothing is streamed. **Trigger** pendant plugged in, key switch turned and dead-man held, at standstill. **Speed** 5 km/h forward, 3 km/h reverse. **Release or link loss** beyond T_{local} = 200 ms → immediate stop, about 0.85 m from 5 km/h at the emergency rate. **Exit** released at standstill → `SAFE_STOP`. Open items: decision 20.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control

System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated	NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B6	**B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.
SHARED	**SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.
	Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.
⊘	**Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→ HPC-sourced, verified by N1	→ actuator feedback
→ island sensor data	→ state · mode · decision
→ control command into the gate	⇢ optional (§3.5)
⇢ fault · `/diagnostics`	

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B6, manual joystick control without streamed sensing

🟥	**full colour** — named for B6 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)
▨	**half-tone** — on B6's path by §3.1, but not named in §3.3
☐	**faded** — not used by B6. An edge is drawn at the weaker of its two ends. Only B6's own tag is drawn on each box.